STELLAR CYBER®

WINDOWS SERVER SENSOR TRANSCRIPT

L01 - INTRODUCTION 2
L01A-PRES: Introduction (04m:25s) 2
L02 - OVERVIEW 3
L02A-PRES: Overview (12m:15s) 3
L03 - SOFTWARE INSTALLATION 7
L03A-PRES: Software Installation (05m:55s) 7
L03B-DEMO: Software Installation (09m:35s) 9
L04 - AUTHORIZATION 12
L04A-PRES: Authorization (05m:15s) 12
L04B-DEMO: Authorization (06m:35s) 13
L05 - VALIDATE INSTALLATION AND AUTHORIZATION 15
L05A-PRES: Validate Installation and Authorization (03m:50s) 15
L05B-DEMO: Validate Installation and Authorization (07m:35s) 17
L06 - ADVANCED CONFIGURATION AUDIT POLICY 19
L06A-PRES: Advanced Configuration Audit Policy (05m:55s) 19
L06B-DEMO: Advanced Configuration - Audit Policy (05m:10s) 21
L07 - ADVANCED CONFIGURATION SENSOR PROFILE 22
L07A-PRES: Advanced Configuration - Sensor Profile (04m:40s) 22
L07B-DEMO: Advanced Configuration Sensor Profile (10m:45s) 24
L08 - ADVANCED CONFIGURATION SYSMON 27
L08A-PRES: Advanced Configuration Sysmon (06m:10s) 27
L08B-DEMO: Advanced Configuration Sysmon (19m:05s) 29
L09 - ADVANCED CONFIGURATION - FILE INTEGRITY MONITORING (FIM) 33
L09A-PRES: Advanced Configuration - File Integrity Monitoring (FIM) (06m:00s) 33
L09B-DEMO: Advanced Configuration - File Integrity Monitoring (FIM) (07m:35s) 35
L10 - ADVANCED CONFIGURATION WINDOWS CHANNEL LOGGING 37
L10A-PRES: Advanced Configuration - Windows Channel Logging (08m:05s) 37
L10B-DEMO: Advanced Configuration Windows Channel Logging (16m:00s) 39
L11 - TROUBLESHOOTING 43
L11A-PRES: Troubleshooting (08m:00s) 43
L11B-DEMO: Troubleshooting (11m:15s) 45

L01 - INTRODUCTION

L01A-PRES: Introduction (04m:25s)

00:00-01:00
Hi and welcome to the *Stellar Cyber Open XDR Windows Server Sensor* module. I'm David Shirley from the Stellar Cyber Enablement and Training teams. The lessons in this module show how to install, configure, and verify a Stellar Cyber Windows Server sensor deployment. We will cover the software installation on a Windows Server and authorization of the server sensor in Stellar Cyber Data Processor. We will cover advanced topics including Windows audit policy configuration, Sysmon installation, the File Integrity Monitoring feature, and Windows channel logging. Throughout this module there may be new concepts and terms you may not be familiar with, but that's okay and to be expected. The important thing is that you become familiar with the concepts and terms described. Through the Stellar Cyber Open XDR training curriculum, your understanding and application of cybersecurity principles

01:00-02:00
will deepen with each repetition, iteration, and example. As you advance, you're on your way to becoming a cybersecurity superstar. Thank you for your dedication.

The *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is made up of the following lessons:

- *Overview*. In this lesson, we'll take a high-level look at where the Windows Server sensor fits into the Stellar Cyber platform, some of the prerequisites based on different network architectures, and confirm resources required.
- *Software Installation*. This lesson takes you from acquiring the Stellar Cyber software to installing, configuring, and validating the installation on Windows server host.
- *Authorization*, with the server sensor software installed on the host, it's over to the Data Processor's user interface to continue by authorizing the sensor. But first we will create a custom sensor profile to use.

02:00-03:00

- *Validate Installation and Authorization*. In this lesson, we validate the installation and configuration using both the Windows host and the Data Processor's UI. By the end of this lesson, we will see Interflow records from the events on the Windows Server in the Data Processor.

- *Advanced Configuration - Audit Policy*. This is the first of our advanced lessons. Although there are already Windows Server events being ingested as Interflow records on the Data Processor, the default configuration in Windows server can be modified to increase the types of events created.
- *Advanced Configuration - Sensor Profile*. This lesson is the corresponding lesson to the audit policy lesson with the Windows server, but this lesson focuses on the Data Processor's sensor profile configuration. It is this profile that dictates what of the available Windows events are ingested as Interflow records into the Data Processor.

03:00-04:00

- *Advanced Configuration - Sysmon*. This lesson looks at the installation of Microsoft's Sysmon, or system monitor service, that provides detailed information about process creations, network connections, and changes to file creation time on a Windows Server host. We will validate the configuration by confirming the ingestion of Windows relating to Sysmon.
- *Advanced Configuration - FIM*. File Integrity Monitoring is an important element of maintaining the security of Windows Servers. This lesson focuses on enabling the feature invalidating the configuration through testing.
- *Advanced Configuration - Windows Channel Logging*. Windows Servers can perform a variety of roles. This lesson shows how to enable events relating to those roles and enable the corresponding ingestion through the configuration of channels in the sensor profile.

04:00-04:25

- *Troubleshooting*, we sometimes make errors during the installation and configuration of the Windows Server sensor software. This lesson looks at how to identify those errors and rectify some of them. Now let's get going!

L02 - OVERVIEW

L02A-PRES: Overview (12m:15s)

00:00-01:00

The first lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Overview. Overview*. In this lesson, we take a high-level look at where the Windows Server sensor fits into the Stellar Cyber platform, some of the prerequisites based on different network architectures, and confirm resources required.

Alright. Let's start with our big picture overview. We'll start on the top left is our Modular Sensor. Our Modular Sensor has a number of features that can be enabled: Log Forwarder, Network Traffic, IDS, Malware Sandbox, Aggregator feature. We'll talk about that a little bit

more and we'll actually do a lesson on that later in this module. The Tenable Nessus scanner is covered in detail in another lesson in a different module. We'll talk about connectors and buffering.

01:00-02:00

On the right-hand side is our Stellar Cyber Open XDR platform. That platform could be either on-prem in a virtualized environment could be on-prem in a Stellar Cyber appliance, or it could be in the cloud your private cloud either hosted in Azure or AWS or OCI, or it could be hosted in our SaaS environment, in Stellar Cyber's SaaS environment, where Stellar Cyber DevOps manages the deployment there. What we want to talk about particularly in this module is our Server sensors and particularly our Windows Server sensor. There's two types of server sensors: There's a Windows Server sensor and a Linux Server sensor. We're going to talk about all the different ways to configure that. Different modules and different features we can enable on that. We'll go through and see how to configure, install, and verify that is working.

02:00-03:00

Windows Server sensors are Stellar Cyber software installed on physical or virtual servers on-prem or in the cloud to collect data used to detect threats on Windows servers. Windows Server sensors collect events within Windows servers from which they create Interflow records and send them to the Stellar Cyber platform. Windows Server sensors collect the following information: hardware events, security events, system events, Windows firewall events, Windows defender events, PowerShell events, and File Integrity Monitoring events, if enabled. We are going to enable that feature in a later lesson. We're about to see what that looks like in a demonstration as well. Windows Server sensor normalizes and converts the collected data to Interflow records and forwards to the Stellar Cyber platform.

03:00-04:00

When the Stellar Cyber platform receives Interflow records from the Windows Server and Linux Server sensors, the platform uses machine learning, statistical analysis rules, heuristic rules, custom ATH playbooks (or custom automated threat hunting playbooks), and Graph ML (or Graph Machine Learning) to identify security threats, generate Alerts, and then correlate those Alerts into Cases for management by our SOC analysts. Here are some prerequisites for our Windows Server sensor. Number one: it has to be run-in at least Windows Server 2008 R2 64 bit and later. Have a Xeon core two virtual calls two gig or more, eight gig of RAM, and 128 gig of SSD disk. We always recommend updating the Windows server software and patch it before installing the Stellar Cyber Windows Server sensor.

04:00-05:00

There's some firewall ports that need to be configured as a prerequisite. We'll talk about those in a moment. There's a couple of tables that we can look at to see what ports need to be open, very similar to our Modular Sensor table. From an anti-virus and EDR endpoint detection response scanning perspective, we need you to white list a couple of folders. They are c:\programdata\StellarCyber and c:\program files/aella. Both of those would need to be whitelisted to stop false positives from being generated from your AV or EDR scanning. Part of the Windows Server sensor software has a number of services that end up running on that Windows host. There's a couple of pages of it here. I think there's five services that run. The first one at the top there we can see is the Windows Agent Sensor Conf. That's the configuration. That's what handles the configuration that is pushed down from the Data Processor when we make changes to the sensor profile.

05:00-06:00

We don't actually turn features on and off for the sensor. We turn features on and off in a sensor profile and we bind the sensor to that profile. When that profile changes, if there's adds, changes, and deletes, we make to that profile. It is this service that handles that configuration change. The Windows Agent Sensor Control, that's what monitors the other services and can start and stop them based on the configuration. The Windows Agent Sensor Phonehome that sends Windows sensor logs to the DP. There were six of them, not five. The Windows Agent Sensor Logbeat sends the Windows events to the Data Processors. We talked about these Windows events will be normalized and then sent as Interflow records to our Data Processor.

06:00-07:00

The Windows Agent sends a filebeat that sends logs from the Windows DHCP server if we have the DHCP server installed and running. If it has those, that is the service responsible of sending those logs to the Data Processor. The final one is the Windows Agent Sensor FIM, or File Integrity Monitoring. We'll be doing a lesson on that later. That's the software that monitors our files and directories for adds, changes, and deletes. There are a couple of different network architectures that we can deploy in. This is the simplest one. Actually, probably the more simple one is to have no firewall at all, But if we have a firewall sitting between our Windows Server sensor and the Data Processor, we require that firewall to have some open ports. Now typically a lot of deployments would have just any any any from inside to outside. But let's just say that we want to be very specific about what we allow out from our Windows Server sensor

07:00-08:00

to the right-hand side of this network, to our Data Processor. There is a list of protocols and port numbers on this page, but it'd be a bit easier if we look at that in a table format. Just remember this is option number one or diagram number one where we have the Windows

Server sensor communicating directly to the Data Processor. A table for that. We can see in the left-hand column Windows Server sensor. That next column DL is for Data Lake. It needs 6640 through 6648 open for TCP with TLS 1.2. That's for communicating with the CM, the configuration manager, to the DL 8443 using TCP (HTTPS with TLS 1.2). That's for downloading software and files from the DP. We could be upgrading our Modular Sensor or our Windows Server or our Linux Server sensor and that's the port that that will happen.

08:00-09:00

Then our custom log parsers will be also pushed to our server sensor via that port number and protocol. The final one is to the DA the Data Analyzer on port 8888 and 8889 TCP (HTTPS with TLS 1.2) and that's the receiver ports for communicating with the Data Analyzer. A second network diagram has some extra features in here. We can see we have now got two firewalls and sitting in between those two firewalls, we have a Modular Sensor with Aggregator feature turned on. What that enables us to do is instead of communicating our Windows Server sensor communicating directly with the Data Processor, we can set up the Windows Server sensor to communicate with the Modular Sensor as an Aggregator, then the Modular Sensor with the Aggregator feature turned on communicates with the Data Processor.

09:00-10:00

That gives us a couple of advantages. If you were to imagine on the left-hand side we've got many Windows Server sensors and many Linux Server and many Modular Sensors and they all need to communicate with that Data Processor. Those firewalls, that firewall on the right-hand side, would need to have all of the ports and protocols open for each of those individual servers–Linux or Windows–and Modular Sensors to communicate with the Data Processor. When we use a Modular Sensor in between, we're able to have the servers, Linux and Windows, communicate to the Aggregator, the Modular Sensor in the middle. The only ports required on that right-hand firewall are the ports required from the Modular Sensor to the Data Processor. That's with the Aggregator feature turned on. The network configuration we're going to go through in our slideware and in our demonstrations actually has that Windows Server and that Modular Sensor on the same network.

10:00-11:00

In fact, the firewall on the left-hand side is not open, but for the purpose of this description we want to show you what ports you'd have to have open on the firewall on the left and what ports you'd have to have open on the firewall on the right. The second question is pretty easy. I'm going to go back a slide. They're the ports that would have to be open. It would say Modular Sensor to the DL, Modular Sensor to the DL, and Modular Sensor to the DA is what we'd have to have open. The firewall on the right-hand side would require this. The Windows Server sensor to a Modular Sensor with the Aggregator feature turned on. You

can see they are all the same port numbers that were required when we were communicating directly with the Data Processor. We now need those open to communicate with the Modular Sensor with the Aggregator feature turned on:

11:00-12:15
6640 through to 6648 and 8443 and 8888 and 8889. We'll proxy those from our Server sensor software to our Modular Sensor and then to the Data Processor. Couple of extra firewall ports depending on your network layout. You may well have an NTP server on the same network as your Windows Server sensor, in which case you don't need to have a firewall rule. But if you had an external one, then we needed NTP on 123, with UDP. And of course then depending on what your DNS environment is we'd need port 53 and UDP open to allow our Windows Server sensor software to communicate both with that DNS server and an NTP server. Very important in a security environment to have all of our security and networking devices synced from a clock perspective.

The lesson overview is complete. Thank you.

L03 - SOFTWARE INSTALLATION

L03A-PRES: Software Installation (05m:55s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Installation and Configuration* module is *Software Installation. Software Installation*. In this lesson we describe how to install the Windows Server sensor software.

The first thing we need to do to install our Windows Server sensor software is to obtain a copy of it. To do that, browse to the Stellar Cyber Knowledge Base at docs.stellarcyber.ai, and then navigate or search. In this example, we're going to navigate to **INSTALLING**, and then **Installing a Server Sensor**, and finally **Installing a Windows Server Sensor**. Then scroll down until you find that obtaining the Windows Server installer. There are two choices there, one with a sha1 validation.

01:00-02:00
After clicking on one of those links, you will then be asked to enter your username and password to download that software. Please contact Stellar Cyber support at support@stellarcyber.ai for those login credentials. Confirm that your software has been downloaded in your browser and navigate to your **Downloads** folder or whatever folder it is that you've downloaded that software into. We can see there we have our software 4.3.7 is

the version we're using. Double click to start the wizard on the downloaded file. Select **Next** to continue. Next step is ask what folder they would like it in. The only supported path is c:\ProgramFiles\aella. If you try to change it you will get an error message. Select **Next** to continue. We've got a couple of different networking configurations that we talked about previously.

02:00-03:00

In this network configuration. This is our Windows Server sensor software communicating directly to the Data Processor. There is no Aggregator feature in between. What we need to do is enter a CM address, or a configuration manager address here. This is the management IP address or fully qualified domain name for your Data Processor's Data Lake or the management interface is how you access your Data Processor. If you don't enter a tenant ID, your server sensor will be bound to the root tenant. If we want to have our Windows Server sensor software bound to a particular tenant, we can see there go to the Data Processor UI navigate to **System** and then **Tenants**. In that table, there is a column for ID. If there's not, just add that column. Right click and copy that ID and paste it in here.

03:00-04:00

When we go to authorize this sensor, it will already be in the correct tenant. If we leave this blank it will simply go to the root tenant. If it's not too late to change that while we're authorizing it or even after we can edit the sensor and change the tenant that it should be bound to. For Primary Aggregator and Secondary Aggregator in this network scenario where our server sensors communicating directly to our Data Processor will leave those blank. In this network architecture where we have communication from our Windows Server sensor software to a Modular Sensor with the Aggregator feature turned on, we still enter a CM address there. You can still choose to put a tenant ID there, but the change in this where we use a Modular Sensor with an Aggregator feature is that we enter a Primary Aggregator and a Secondary. If you don't have a secondary, that's okay. You can only have one that will work.

04:00-05:00

If you don't enter a primary and you do enter a secondary, it will accept it, but it won't work. If you've only got one, enter it as a primary. It's not like we can have both. We can't have it communicating directly to the configuration manager or directly to our Data Processor and a Modular Sensor with an Aggregator feature. It's one configuration or the other. This is where we would type that in during this configuration. Then we choose **Next** to complete the install, and then select **Finish**. To validate the installation of our Windows Server sensor software, click the **Windows** button. You'll see down the bottom there under Windows Agent Sensor. There is a Windows Agent Sensor CLI. You could also just click the Windows key and

type in CLI, and this will show up in that search result. Let's select that. We'll see that opens up a CLI.

05:00-05:55
It looks very similar to our console of a Modular Sensor. There are a couple of show commands in there a little differently, but it's similar. The first thing we want to do is do a `show cm` or `show configuration manager` and we want to see that our IP address is correct and that the status is up. We now have this Windows Server sensor software communicating with our Data Processor. So far, so good. If you get a "Fail to communicate with the CLI service," if that's the result, if that's the output that comes up after you enter `show cm`, just wait a few moments while it's communicating with your Data Processor and just run the `show cm` command again.

The lesson, *Software Installation,* is complete. Thank you.

L03B-DEMO: Software Installation (09m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Demonstration of Software Installation. Demonstration of Software Installation.* In this lesson, we demonstrate how to install the Windows Server sensor software.

In this demonstration, I will show you where and how to download the Windows Server sensor software, install the sensor software on a Windows Server host, and confirm the installation of the Windows sensor software. Let's start with where and how to download the Windows Server Center software. Let's first go to our documentation. We'll open up a browser and it is docs.stellarcyber.ai. Here we are at the home of our Knowledge Base.

01:00-02:00
We can do this a couple of ways. We can either use this left-hand side to browse for the page we're looking for or we could use the search tool up in the right-hand corner in this search bar to search what we're looking for. Both of them take us to the same place. Let's do the browse because I'm sure everyone knows how to do a search. We're installing so I'm going to click on the drop down of **INSTALLING**. We're installing a server sensor. I'll choose the drop down on that. We are installing a Windows Server sensor. That's what I'm selecting. Here we are in the documentation on how to install our Server sensor. Couple of things for us to look at: the processes that will run will confirm that these processes are installed on the server after we have installed our server sensor software.

02:00-03:00

A recommendation from an antivirus and EDR, or endpoint detection and response, scanning to exclude those directories from your scanners whether you've got antivirus or some sort of EDR. Sysmon we will talk about later in this module. **Supported Systems**: We can see here that Windows is supported on 2008 R2 and later. Let's have a look at what we're running. We'll go and open our server manager, and tools, and system information. We can see here we're all good. We're running Windows Server 2019 so certainly more than 2008 standard R2. What do we need? We need a Xeon Core 2 processor and eight gig of RAM and 128 gig of disk. Let's see what we're running. We've got four CPUs, so we're okay there.

03:00-04:00

From a memory perspective we're good. We've got sixteen gig of RAM. From a disk space perspective, let's see what we've got. This machine has 128, so we're all good there. Back to our documentation. We are all good for what we need to do. Of course make sure you have updated your server with the latest patches to do there. Firewall requirements: We covered that off in the previous lesson. Here we are to download our server software. The software can be downloaded from the production server directly by using the URLs. We're just going to choose the first one. You can see here actually at the top we are running, we are looking at the documentation for 4.3.7. If you had a different version of Stellar Cyber software that you're running, we're going to upgrade this one to 5.1.1. This is the reason we're going to install 4.3.7.

04:00-05:00

We can see what it's like to upgrade. We'll see the process to upgrade from a 4.3.7 sensor to a 5.1.1 sensor. At this stage, we're going to download this. It says here, "Installation link if you wanted a different version, simply substitute the version that you're looking for." We could use this link and instead of 4.3.7, you could put 5.1.1 and you would get the latest version or whatever version that's available. Let's just start with clicking that link. We're going to be asked for a username and password to download this software. This is available from our Support at stellarcyber.ai. We'll pass on that information to you. Enter your username and password that you're given by Support and sign in. Just about down a hundred meg of software and select from in there to start the process. We could go have a look and see where it was downloaded to our **Downloads** directory, and there is our piece of software.

05:00-06:00

Let's double-click here to start the installation process. Okay. Here's our wizard to start the installation process. Select **Next** to continue. The folder to install this software must be in the Aella program files. Aella: Don't change that. There'll be an error if you try to change that. Simply select **Next**. Now we have a couple of choices in how we fill this out depending on

what your network layout looks like. If you have a network layout where your Windows Server will talk directly to the Data Processor, we would type in the address here. Our Data Processor is 192.168.1.175. That's what we would type in. Now if we needed a tenant ID, if that's left blank, this Server sensor will be bound to the root tenant.

06:00-07:00

If we wanted it bound to a particular tenant, we'd need the **Tenant ID**. The way we find the **Tenant ID** if we go to our DP and we go **System** and **Tenants** we can see here we've got a couple of tenants. We've got the root tenant, and we have *User Customer One* and *User Customer Two*. But in this list, in this table we cannot see the **Tenant ID**. Let's click **Columns** and scroll to the bottom and there is the ID field. Now we have an ID. You can see this is a really long number. You just right-click, copy to clipboard, and go put that in pasted into our **Tenant ID**. That would bind it directly to the tenant. We want to put this one in the root tenant so I'm going to remove that. Next, we can see you are ready to install.

07:00-08:00

Select **Install** to continue. Let's wait for this wizard to continue through. You can see that it's doing an install. It will also start the services required, so you'll see a bunch of files being copied and then you'll see some services that will be started. There's the services being started. Now the copy is finished. The wizard has finished. Select **Finish**. Alright. Let's go and see. First, for some confirmations what we could do is to see this. We saw in our documentation earlier that there were a number of processes that are going to be running, scroll to the top and there is our list of processes. Let's just go make sure those services are running.

08:00-09:00

We could go back to our **Server Manager**, **Tools**, **Services**. I know they start with "w," so I can click on one of these and press "w" to get towards the bottom. Here we are, Windows Agent Sensor Conf Windows Agent Sensor CTL, Filebeat. FIM. We'll talk about file File Integrity Monitoring later in this module. Logbeat and a Phonehome. We can see they've all been installed, which is fantastic. That is not installed, but installed and running. The next piece for us to do is to go and see from the start menu, Windows Agent Sensor, and we can see there is a Windows Agent Sensor CLI. Let's run that. It's going to look a little similar to the CLI of our Modular Sensor. We could do a `show cm`.

09:00-09:35

The good news here is that we've configured it correctly. We have a 192.168.1.175 for our CM controller. The status is up. That's the most important thing there. The next thing for us to do is to go into the Stellar Cyber user interface on our Data Processor and finish the configuration there.

The lesson, *Demonstration of Software Installation*, is complete. Thank you.

L04 - AUTHORIZATION

L04A-PRES: Authorization (05m:15s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Authorization*. In this lesson, you will learn to authorize the sensor in the Data Processor.

Now that we have downloaded the Windows Server sensor software and installed it on our Windows Server host, it's now time to authorize that sensor in the Data Processor. Sign in to the Data Processor and navigate to **System**, and then **Sensors**. We can see here that our sensor has appeared in our sensor table, our Windows Server sensor AD3SRV, that is the host name of our Windows Server. Notice also that the **Sensor Status** is green, so that's great news. That's the equivalent on our Windows Server sensor our CLI of doing a `show cm`

01:00-02:00
and seeing that status of up, that gives us our green light at this end. But the authorization has a red shield, so it's not authorized. That's the process that we're about to go through now. The only other thing I want you to pay attention to is by default, when we have a new sensor arrive at our Data Processor, it assigns that the default profile. For a Windows Server sensor and a Linux Server sensor, it will get the default profile. For a Modular Sensor, it will get default modular. Stellar Cyber recommends using a customized sensor profile rather than the default sensor profile. Depending on what your network configuration requires, we advise you to create a customized sensor profile for your environment, your specific use cases. To do that, we navigate to **System** and **Sensor Profiles**.

02:00-03:00
Then we need to create a standard one. We can see the two default profiles are there: default and default modular, but we're going to create a new profile for our Windows Server sensor. Click the **Create** button. From that drop down, we're going to add a standard sensor profile. Select **Add Standard Sensor Profile**. The window that opens up allows us to add some configuration settings to that sensor profile. The first thing we're going to do is give it a name, choose a receiver. Just use the default receiver unless you've configured your network in a different way, but for this example we're going to use the default receiver, and

then select the **Submit** button to create that. Now note there are a lot of features in there. You can see there's a lot of tabs and a lot of slider bars for features. We're going to cover those features in another lesson. Although in this module we will be covering off a number of the Windows Server sensor profile features.

03:00-04:00
Now we have configured our standard profile called, "Training Server Sensor Profile" that's been created, but all we've done is create the profile. We now need to bind our new Windows Server sensor that's appeared in our sensor table list and bind it to this profile. To do that, navigate to **System** and **Sensors**, edit the server sensor by clicking the pencil icon in the table, and it will open up the sensor parameters to edit. We can see there, to edit those there's a drop down next to the sensor profile. In that drop down we can see our new standard sensor profile. Select that and then choose **Submit**. The last thing to do for us is to authorize this sensor. Now we can see that the status is still green. **Authorization Status** is not authorized,

04:00-05:15
but you can see that we have a sensor profile changed from default to our Training Server Sensor Profile. To authorize this, select **Manage**, and then **Authorization** from that drop-down menu. Next is to authorize our sensor. Starting the top we can see we have **Authorize** and **Unauthorize**. We've got the radio button selected to **Authorize**. In the left-hand panel, we can see our server that's there ready. Select the server, and then click **Add** to add it to the right-hand panel of those ready to be authorized. We can now see it in the right-hand panel. Next is to select **Submit**. Note the Windows Server sensor software will now restart not the Windows Server, just the Stellar Cyber Windows Server software will restart as a result of pushing this authorization and pushing our server sensor profile down to that Windows host. Thank you.

L04B-DEMO: Authorization (06m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Demonstration of Authorization Demonstration of Authorization.* In this lesson, the authorization will be demonstrated for Windows Server sensor software.

In this demonstration, I will show you how to authorize the Stellar Cyber Windows Server sensor in the Data Processor's user interface. Let's start with the DP's user interface. To see new sensors in our system we're going to navigate to **System** and then **Sensors**, and we

can see a table here. The table on the right-hand side is the sensor name. That name comes from the server's name. Let's go see if we can see what that is.

01:00-02:00

We'll go to local servers in the server manager and we can see here our computer name is AD3SRV. Now if we go back to our UI, we can see here AD3SRV that's our server sensor. The good news is we've got a green light for **Sensor Status**. What we don't have is under our **Authorization** column it is not authorized. So we need to do something about that. Now something else to notice here is the sensor profile that it's using. If we expand that column out is default. Now we recommend that you do not use the default uh sensor profiles to create your own one. So let's go and create a sensor profile for all of our in our network as it grows and changes, all of our service sensors. To do that it's a pretty simple process, **System | Sensor Profiles**.

02:00-03:00

We can see we've got a bunch here already. Let's create our own one. Because it's a sensor profile it is a standard sensor profile. Let's give it a name. Let's give it something like, "Training Server Sensor Profile." We need a receiver. We'll choose the default receiver and these are the defaults Let's just leave all those defaults as they are right now and submit. We can now see we have Training Server Sensor Profile. It is a standard profile type so it's ready for our server sensors. Back to our sensor list: **System** and then **Sensors**, here is our list. If we edit this one We can click on it to look at it to see what it's doing and we can see its name, its sensor ID, its software version, what features it's running, the status of what features it's running.

03:00-04:00

But most importantly here the **SENSOR PROFILE**, we could click on default there would be one way of doing it. That will edit the profile, but that's the default profile, We don't want that. Let's go back to our table, **System | Sensors**. We want to edit this sensor. Select the pencil, and we can see here it's in our root tenant. This would be the other place to do it. If you didn't when configuring or installing the server sensor software put in your **Tenant ID**. This is the other place we could go and change it. Once it's been bound to our root, we can see up here we are in **All Tenants**. In our root tenant, is where this sensor was bound. Here is the IP address of that host, the host name, a DNS name. It doesn't have a DNS name but we're coming down to sensor profile and we want to choose Training Server Sensor Profile, and submit.

04:00-05:00

We have our server that is still not authorized, but it is up and running and it's up and running because if we go to our sensor software over here we can do a `show cm` because

this is up and running. That means they're communicating, it's just not authorized. Back to here. Next thing to do is you can either if you were doing many of them you could click these for multiple tasks but not important to do the checks there. We can just go up to **Manage** and then **Authorization**. We can see here we only have in our list to authorize one sensor to authorize. We'd see Modular Sensors here, Linux Server sensors, and of course we're seeing our Windows Server sensor there. Let's select it. Select **Add** to bring it across, We can now see it. We're authorizing it. We're not unauthorizing them. And we can now see that has been added to our list. It's just a matter of submit.

05:00-06:00
Our server sensor has been authorized. We can see it authorized here, but we have a big fat red dot. It's disconnected. Just wait a few moments. You can hit this refresh button up here. Refresh. It's still not up and running. We could jump into our server. Let's see how we're going in here. (types) `show cm`. We're still waiting for that to come up. When we click that **Authorize** and **Submit** it actually will reboot the software not the server, the software that's running here. We're just waiting for that `show cm`. Wait a few moments while that comes up. We'll just check again. Alright. We see a `show cm` there of our 175 and it's up. Go back to our UI of our DP and refresh.

06:00-06:35
We've got a green button. We can see that our server is up and running. We can see our server with the server sensor software running on AD3SRV has been authorized and the **Sensor Status** is Success.

The lesson, *Demonstration of Authorization,* is complete. You should now be familiar with the authorization process for Windows Server Center software. Thank you.

L05 - VALIDATE INSTALLATION AND AUTHORIZATION

L05A-PRES: Validate Installation and Authorization (03m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Validate Installation and Authorization. Validate Installation and Authorization.* In this lesson you will learn how to verify that your software installation and the authorization are properly installed and operating correctly.

We have now installed, well downloaded and installed our server software on our Windows host. We have authorized it in the Stellar Cyber platform. We've also given it a new server sensor profile. Let's validate that all we did or that configuration has all worked. The first thing we can do is go to our Windows host and run that Windows Agent Sensor CLI app. We can see once we're in there we can do a `show cm`.

01:00-02:00

Look and we can see the `show cm` gives us an IP address of our CM, which is our Data Lake Management IP address of 192.168.1.175. The other piece we want to see in there is the **Status** of "UP." From the Windows host perspective the configuration is working. If we go into our Data Processor UI, let's have a look at what we see in there. If we go to **System** and then **Sensors**, here's a couple of things we want to observe. First is the **Sensor Status**. A green dot is what we're looking for from a status that syncs up with the status of UP there. Those two things are combined to give us that. If we then look at our authorization we've now moved from having our red shield not authorized to green shield and authorized. That is good.

02:00-03:00

Next thing we'll do let's actually select the sensor host name to see another view. In this view, we can see the **SERVICE STATUS** panel to see which of our services have a green status. We'll notice there that we have File Integrity Monitoring is red. We have not turned that on yet. We will certainly do that in a later lesson. We can see our sensor profile here is the profile will change from default to the profile that we created. Final confirmation, let's go look for some ingested events. To do that we'll navigate to **Investigate** and **Threat Hunting**. We know when we're going to **Investigate | Threat Hunting**, the default index is **Alert**. We need to change from the **Alerts** index to the **Windows Events** index. Uncheck **Alerts** and check **Windows Events**. What we can see here now so confirm what we're looking at is **Top Sensors** and **Top Log Sources**.

03:00-03:50

We can see that our sensor is there, our host name of the sensor, and we can see some Windows logs. If you look further down in the table, we can see that there are some event IDs. We've added that **Event ID** column to that table and we can start to see the ingestion of Windows events and the actual event ID in the system. We've downloaded the software, installed the software, authorized the sensor, and now we've checked to see that it is working.

The lesson, *Validate Installation and Authorization,* is complete. You should now be familiar with how to verify software installation and authorization for Windows Server sensor. Thank you.

STELLAR CYBER®

L05B-DEMO: Validate Installation and Authorization (07m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Installation and Configuration* module is *Demonstration of Validate Installation and Authorization. Demonstration of Validate Installation and Authorization*. This lesson will demonstrate how to verify that your software installation and the authorization are properly installed and operating correctly.

In this demonstration, I will show you how to validate the installation and the authorization of the Stellar Cyber Windows Server sensor in both the Windows Server Host and in the Stellar Cyber Data Processor's user interface. Let's start with the Windows Server. In our server we can run our Windows Sensor CLI.

01:00-02:00
We know in there once we get a prompt, we can do a `show cm`. What we're looking for is our controller, that CM controller IP address 192.168.1.175. In this configuration, the server sensor is communicating directly to the Data Processor. There is no Aggregator in between. We can see there that the **Status** is UP. From the Data Processor's user interface, here's what we can see. Let's say **System | Sensors**, and we've got a table of sensors. We know our sensor is its host name. We can just stretch that out. The server name is the host name AD3SRV. We can see we have a **Sensor Status** of its green success, and we are authorized. If we select that sensor, we can come in here and see its name. Its sensor ID, the software version it's running, that it's a Windows Agent Sensor,

02:00-03:00
which profile it is using. We're using the *Training Server Sensor Profile*. What the IP address is of the host. Over here we can see what services and the status of those services. We can see the configuration's good, control's good. **File Integrity Monitoring** is red, and we'll see why. We haven't turned that on in the profile, which we will do in a later lesson. **Windows Log Events** is green and **Diagnostic** is green. All that looks good. Let's go in and see what other or how else we can confirm that our server sensor software is working. Let's go to **Investigate** and **Threat Hunting** to see what is being ingested. We know when we go to **Investigate | Threat Hunting**, that the indices defaults to **Alerts**. We want to go look at Windows Events, so we select the **Alerts** button,

03:00-04:00
uncheck **Alerts**, scroll to the bottom and select **Windows Events**. What we can see here now is under our **Top Sensors**, we now have an AD3 server. Let's filter for. If we highlight that there's a plus and a minus here. We're going to click the plus, which is filter for, meaning I

only want to see Windows Events because that's the indices we're looking at. I only want to see Windows Events that are from this sensor, AD3SRV. I click the plus button there. We can see here it's added a filter, AD3. That's the only one we're looking at. Click the **Filter** button to hide that column. We can see we have in the last 24 hours we've only just turned this on we have 11 entries. We can see over here that in the legend we've got Service Control Manager as a class. We have Microsoft Windows Security Auditing,

04:00-05:00

Microsoft Windows Security SPP. Now if I go down to the table to see here are the 11 entries, I want to see what is the event number? The way to do that is if I expand one of these records, I can see all of the fields inside of that record. Now I can see this record has an **Event ID** of 7036. When I highlight that field there I get a pop up of these icons to choose what I can do. One of them is **Show column**. I can filter for. I can filter out. I can copy to the clipboard our 7036 or I can copy this as a query if I wanted. All I'm looking to do is to add the column. The quickest way to add that **Event ID** column is to click the **Show column**. We can see now if I minimize that I have an **Event ID** column. Here are a bunch of records for our server, from our server software,

05:00-06:00

sensor software that is showing us a bunch of different Event IDs. Let's open up one of these records and see what we can see in here. Let's do the 4625. If we come over here into the actions column there's More Info with the little "**i**" button. We're going to select that and now we get a panel. The first panel we see is the **Key Fields**. They're the **Key Fields** for this record. Not too many fields in there, but key ones. Let's go into the **Details** and just see what we can see about this server sensor software running on our Windows host. Number one: We can see that, well this app name was a Windows login. That's got more to do with the record that's come in. We can see the computer name. That's the full name for that record. Destination IP address was a dot three for this. Let's scroll down and see the other pieces of information that are interesting to us.

06:00-07:00

The engine device class is Windows. We can see the engine ID name. That's the name of our sensor AD3SRV. Here's where we see our Event ID of 4625 and our host IP address of our 10.20.1.13. The message origin source of Windows_agent. That's what shows up in our **Top Log Sources** in the pie chart above. This confirms that our Windows Server sensor software is working on our server AD3SRV. What I would say though is it's been a little bit time since we've run this. We're looking in the last 24 hours, but we've only I know we only just created this and we'd only just authorized it. I'd expect to see so many more events. We know that Windows is very chatty.

07:00-07:35
Let's continue on and see how we can increase the number of classes that we would expect to see here and certainly go up from 11 to a much more significant number of events coming from that server.

The lesson, *Demonstration of Validate Installation and Authorization,* is complete. You should now be familiar with how to validate installation and authorization of the software. Thank you.

L06 - ADVANCED CONFIGURATION AUDIT POLICY

L06A-PRES: Advanced Configuration Audit Policy (05m:55s)

00:00-01:00
The next lesson in the *Stellar Cyber open XDR Windows Server Sensor Installation and Configuration* module is *Advanced Configuration Audit Policy. Advanced Configuration Audit Policy.* In this lesson, you will learn how to expand the types and number of Windows events collected by the Windows Server sensor and forwarded to the Data Processor.

Up until now we have configured our server sensor software on our Windows host. We can already see that some events are being ingested into the Stellar Cyber DP. We just want to make sure that as many as we'd like to see are being ingested. We can see here from a Microsoft document looking at the following table, by default, Windows has the categories and subcategories policy for success and failure set to no and no.

01:00-02:00
It's got that set for Account Logon, Account Management, Detailed Tracking, and DS access. Not really as much as we like. Stellar Cyber suggests using the Stronger Recommendation setting. Now that setting is in the right-hand column. That is a Microsoft recommendation of what gets turned on and off on our Windows host. To change that, there's two ways we can go and look and see what's configured in our Windows host. The first is via the command prompt `command` and the second is to use the local security policy app. Let's look at the first one through the command line. Open up a command prompt, run as administrator, and issue the `auditpol /get /category:*`. The `/get` displays the current audit policy for more information. You can go to see audit policy get for the syntax options, and the `/category:*` says we want to see all of the categories.

02:00-03:00

The big note down at the bottom, which is pretty important, if it says “Not configured,” from Stellar Cyber's perspective, means none of those Windows events will be sent to Stellar Cyber for us to act from a security perspective in generating Alerts and correlate into Cases. The other way to view this audit policy is to look at the **Local Security Policy** app. Remember we're just looking at this from a single server. If you've got a Windows domain with Active Directory, you can look at global policies as well. We're just looking at the local policies in this example. We open up the **Local Security Policy** app, expand **Advanced Audit Policy Configuration**, and then select **System Audit Policies**.

03:00-04:00

We can see there they are all not configured. We're not going to receive any Windows events relating to those categories. Now how do we go about turning those on? If we stay in this local security policy app, we expand **System Audit Policies** and select the account log on. We've got to do these individually. We can't do these as a group. The first thing we do is select the first one Account logon and then can't really see this because the pop up in the screen capture’s hiding it but behind it on the right-hand side there you can see that all of those subcategories have been highlighted. You can just sort of click in that space, control+a to highlight them all, and then right-click on properties for each of those that are under Account logon. That's when we get this pop-up window. From the pop-up window, we want to select, **Configure the selected events to be audited**.

04:00-05:00

That's the first check. All of these are unchecked when you first get this pop-up window. Then select **Audit all success** and **Audit all failures**, and then click **OK**. Now you're going to have to repeat that same process for the rest of those categories. You're going to have to do it for Account Management, for Detailed Tracking, for DS Access, Logon/Logoff, Object Access, Policy Change, Privilege Use, and System. The bottom one we won't do because it requires more configuration to actually create policies rather than just check and turn them on. Go ahead and do all of those. Remember you click on the subcategory, control+a to highlight them all right-click for properties, and then we're going to turn on **Configure the selected events to be audited**, and we want to audit for all successes and all failures. When you finish doing that, click **OK**.

05:00-05:55

If we go back to our command prompt now and again run `CMD` as an administrator. This time we do an audit pol /get /category:* we can see we've had a great change there. Now in some instances depending on your Windows environment and the configuration, you might have to do a `gpupdate` with a `force` command, but just check to see what your environment looks like whether that `gpupdate` is a requirement or not. A `gpupdate` is a

global policy update. If you are using global policies across multiple servers, if you're using it on a single server, there is no need to do this command.

The lesson, *Advanced Configuration Audit Policy,* is complete. Thank you.

L06B-DEMO: Advanced Configuration - Audit Policy (05m:10s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Installation and Configuration* module is *Demonstration of Advanced Configuration - Audit Policy. Demonstration of Advanced Configuration - Audit Policy*. This lesson includes a demonstration of how to ban the types and number of Windows events collected by the Windows Server sensor and forward it to the Data Processor.

In this demonstration, I will show you how to expand the types and number of Windows events being collected by the Windows Server sensor and forwarded to the Data Processor. This requires a change of configuration to the Windows Server host. Let's start with looking at the default settings in Microsoft's documentation. Here we are on a web page for Microsoft documentation regarding audit policy recommendations.

01:00-02:00
We can see that this applies to our server, our Windows Server 2019. Let's go see what some of these recommendations are. As an example, we have our **Audit Policy, Category or Subcategory** for Account Logon. By default, we can see there the Windows default for success and failure is set to No and No. There is no configuration set here but yet the recommendation or the stronger recommendation is to turn a lot of these on. We're going to go have a look at our server and see where those are and what changes we can make to increase the breadth of Windows events being sent by our Windows Server sensor. Let's first look in the command line of what we can see if we search for a CMD command prompt. I'm going to right-click here to run as administrator.

02:00-03:00
I'm going to execute the command auditpol /get /category:*. Now we can see here what is being turned on or off. What we're paying particular attention to is “No Auditing” means there's nothing about these going to Stellar Cyber, our Data Processor to make security assessments based on the events that are being forwarded. That's one way to do it via the CLI. Let's look at another way of doing it. Let's close that off. We can go to **Tools**. We're going to look at the **Local Security Policy**. Let's make that bigger. We're going to expand **Advanced Audit Policy Configuration**. That's the bottom one here. Expand there.

03:00-04:00
Choose **System Audit Policies**. Now we can see a bunch here configured, well or not configured as the case may be. We have “Account Logon Not configured,” “Account Management Not configured,” a bunch of these are not configured. Let's go ahead and change these to configure them. The way we do that is expand down here, select that, come in here, highlight all of them. I did that with a control+a, right-click, properties, and we want to turn that on. We want to audit everything, success and failure and click **OK**. Now we have that one. We've got to do this for each one of these. Let me just go ahead and do this. I'll stop talking so we can speed up the video.

04:00-05:10
Now for this final one, this **Global Object Access Auditing**, these require the configuration of defining policy, so if you want to go ahead and do that that's a bit too detailed for what we're trying to achieve here. Same thing for **Registry** right-click properties, and you would have to go ahead and define particular policies in there. We'll just leave that alone for the moment. Now our server is ready to send those additional Windows events to the Stellar Cyber DP.

The lesson Demonstration of Advanced Configuration Audit Policy is complete. Thank you.

L07 - ADVANCED CONFIGURATION SENSOR PROFILE

L07A-PRES: Advanced Configuration - Sensor Profile (04m:40s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Advanced Configuration - Sensor Profile. Advanced Configuration - Sensor Profile.* In this lesson, you will learn how to modify the standard profile for the Windows Server sensor using predefined templates to match common deployment scenarios.

By default, Windows Server sensor is collecting a minimal set of events. Consider using **Set to Template** and choose a Windows compliance profile, which is the high-volume profile. What we can see here is a table that's been taken from our documentation from the Knowledge Base. Previously what we've done, I mean we've got to look at this in two steps. First of all, what is Windows writing events for in into the event log?

01:00-02:00

Two: What is the Stellar Cyber Windows Server sensor collecting from the event log and ingesting into Stellar Cyber? There's a bit of a two-step process. First of all we've configured the Windows Server to produce more events for what we're looking for. The second piece here is to tell the server sensor via the profile what to collect. Here's the example. We have a table here where we can see we've got the Windows default profile, the context profile, and the compliance profile. Basically it's low, medium, and high. Let's have a look through how we go ahead and configure this. For a list, go to the documentation in the Knowledge Base and you can see a very long list in this table. To change the configuration, remember we don't change the sensor itself, we change the profile that the sensor is bound to. In the UI, we select **System** and then **Sensor Profiles**.

02:00-03:00
Select the pencil icon next to the sensor profile you want to change. In this example, it's the Training Server Sensor Profile. First thing we're going to look at is the **Windows** tab. In this standard profile, we'll check the **Windows** tab, select the **Windows** tab. Then, just as an example, we can see at the top there it says, "View the current settings related to Windows Logon/Logoff Events. If we expand the **Security** section, we can now see the Logon/Logoff events. One of those is checked. At the moment it's minimal. Observe the events that are not selected in the profile. What we're going to do now is go change that profile and come back and look to see what is checked. Of course, you can choose to do this entirely manually, but Stellar Cyber's given you three templates:

03:00-04:00
low, a medium, and a high to select that we'll choose to turn on and off selected events. To make that change, in this example we're going to change it to a Windows compliance profile or the high-volume profile. We're going to select **Template** from up the top. We're going to select **Restore Windows Profile to Template**. You can see we have three choices there: We've got the low, the medium, and the high or **Detect**, **Context**, and **Compliance**. We're going to select **Compliance**, the high-volume one. Then select **Yes, overwrite** to overwrite. Select that last, and now we've overwritten it. Let's go and check again. To view the Windows Logon/Logoff events, click the **Windows** tab, expand the Logon/Logoff events, and we can see now they're all selected. That was the profile that made that change.

04:00-04:40
As I said before, you are welcome to manually select and unselect any of these events you wish to suit the environment that you're working with. At the end of that, select **Submit** to save. The configuration of that profile will be pushed down to any sensor that is using that profile. In our example, it'll be sent down to that Windows Server sensor profile and that configuration will change to say, "I want you to start ingesting all of those additional ones we can see checked here."

The lesson *Advanced Configuration Sensor Profile* is complete. Thank you.

L07B-DEMO: Advanced Configuration Sensor Profile (10m:45s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Demonstration of Advanced Configuration - Sensor Profile, Demonstration of Advanced Configuration - Sensor Profile*. This lesson includes a demonstration of how to modify the standard profile for the Windows Server sensor using predefined templates to match common deployment scenarios.

In this demonstration, I will show you how to modify the standard profile for the Windows Server sensor using predefined templates to match common deployment scenarios. Let's start by looking at the Stellar Cyber documentation showing the tables identifying the different templates. Let's jump into our Knowledge Base. We can see here we are at the home screen for it.

01:00-02:00
We've previously used that left-hand panel to browse to get to where we're going, but let's use the search. Let's search for Windows Server sensor. The one we're looking for is the top one: **Templates for Windows Server Sensors**. We'll select that. What we can see here is **Templates for Windows Server Sensors**. Stellar Cyber has provided predefined templates for us. We see here they're called **Windows Detect Profile**, **Windows Context Profile**, and **Windows Compliance Profile**. There's a low volume, a medium volume, and a high volume. The volume is talking about Windows events. If we go look at this table below, we have a Channel. This channel is called **Security**. Inside **Security** there are some categories. In this example, we've got **Account Logon Events**, **Account Management Events**, **Detail Tracking Events**,

02:00-03:00
DS Access Events, **Logon/Logoff Events**. What we can see here is with these green ticks and a little tricky to see with the lime color tick there, but we can see out our columns. **Windows Detect** is our low volume, **Windows Context** is our medium volume, and **Windows Compliance** is our high volume. You can see here in the high-volume column nearly all of them are ticked. Let's pick an example. Let's look at our **Logon/Logoff Events**. We can see here for **Logon Logoff** for UEBA alerts, they're on for all profiles. For **Special Logon**, only available for the medium and the high profile. Let's use that one. Let's look at this one as we turn them on and off. To do that, let's jump into the UI and look at our sensor profile, our standard profile for a Windows sensor.

03:00-04:00

From our UI, the easiest way to find out exactly which sensor profile it is that we need to edit is to select **System**. Let's go straight to the **Sensor** page. We know what our server name is. It's our AD3SRV. Along that line we can see under **Sensor Profile**. Make it larger. It is using the Training Server Sensor Profile. If we click that to view it, it's actually not to view it. It's also to edit it. We are now editing the Training Server Sensor Profile. Of course that's the one that the AD3SRV was bound to. If I go over to the **Windows** tab, we can start to see some of these channels. The first one we're going to look at is the **Security**, and we were talking about Logon/Logoff Events. Now if we go and look at what we've got there, and I'm going to try and fit this all on the screen at the same time, we have three of them turned on.

04:00-05:00

Going back to our documentation, we can see the three turned on is for the medium, for the middle one. Now if I go back to our UI, and let's set the template to restore it to the low volume, the Windows Detect Profile (Low Volume). We select that. It's going to ask us, "Do we want to overwrite the settings?" Basically this is going to make changes to all those checkboxes, select **Yes** to overwrite, and we saw that the special log on, the checkbox for that became unchecked because if we look in our template, for the special log on here, it is only on for medium and high. Let's go back to our UI, and this time we'll change it back to medium, our Windows context profile. Let's keep an eye on our special log on, choose **Yes** to overwrite, and now we have that checked again.

05:00-06:00

Let's go all the way and choose the high volume and where we want everything choose the **Windows Compliance Profile** for high volume and overwrite. Boom! We have a whole bunch of them that get turned on based on the template. Now of course you can go in here and be very detailed about what you want to turn on and what you want to turn off. Maybe under **Security**, **Account Logon Events**, maybe you're not interested in **Other Account Logon Events**. You could turn that off. The template just sets up a default number of events being checked or unchecked. Of course, you can go in and manually overwrite any of them. We can come in here and I could say, "No, no, no, no I don't want any of those" and turn them off, but let's have them all on. Let's go to our high volume and overwrite, and we should see all of those back on. Great!

06:00-07:00

Then the last thing left to do is select **Submit** to make that change. When we select **Submit**, that's made a change to the profile. You can immediately see that our sensor is disconnected. What has happened is it's now pushed a new configuration down to any sensor that is bound or using this sensor profile the Training Server Sensor Profile. Any

sensor, any server that's using that will have its configuration updated because we changed what's being collected in there. We should wait a couple of minutes, and then if we hit this refresh on our table we should see that that sensor has come back up with that new configuration. Any of those events that were being logged on that server, is now from the sense of perspective of how that's been reconfigured, will collect those and forward them on to the Data Processor. We just did a refresh then and we can see that we're green.

07:00-08:00

We're up and running again and we're bringing in new events, new Windows events. Let's go back to **Investigate | Threat Hunting** and see what additional Windows events are coming in. **Investigate | Threat Hunting**. We're going to change away from **Alerts** and we're going to select **Windows Events**. There is our AD server. We're going to filter for. We only want to see in this display only events that are coming from our AD3SRV so field to for. I think we had about 13 events before. We've now got over a thousand events coming in. With that change in the sensor profile, the standard profile for our Windows Server sensor that's been pushed down to the software running on our Windows Server, we can see that there's many more events coming in.

08:00-09:00

Even if we go over to our legend for the different classes, we can see that the number of classes has increased in what we're seeing over here that's already coming through. Many more different types of classes for our Windows events. Let's just look at the Interflow dictionary, and we'll look at Event ID. I could scroll and look for it. It looks like it's going to take a while. To find it I search for Event_ID, and there we have it: Event_ID. Now we can see this is a list of the top ten. Let's expand that out and look at the list of the top 100. We have a whole bunch, a whole host of different event types that are coming into our Data Processor from that server. We still have that filter on.

09:00-10:00

We're still only looking at events coming from AD server three, but we can see that there is a huge volume of event types that are coming in. The nice thing, what this Interflow dictionary does for us is show us the volume of what's coming in. Clearly number one there is 7036 is the largest number of event types coming in. We can filter these out. Let's remove that one. That's a big one. Let's remove that. Then we start to see the next two. 4624 and 4672 are the next most popular event types coming in from that server. This is a really good way for us to see that the we now have a huge increase in the volume and breadth of event types coming from that server from making the change on the server itself of what events are being recorded and also making the change to our sensor profile

10:00-10:45

for what events our Windows Server sensor software will pull from the event log on that server and ingest into Stellar Cyber ready to apply some security analysis. We just close that Interflow dictionary now and here we are back. Let's just remove all of our filters, clear all, and we're back to seeing all of our data coming in. This is just the last 24 hours. We can see there are more servers than just our AD3 over here.

The lesson, *Demonstration of Advanced Configuration - Sensor Profile*, is complete. Thank you.

L08 - ADVANCED CONFIGURATION SYSMON

L08A-PRES: Advanced Configuration Sysmon (06m:10s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Advanced Configuration - Sysmon. Advanced Configuration - Sysmon*. In this lesson, you will learn how to use Microsoft Sysmon system servers to ingest detailed information about process creation network connections and changes to file creation time in addition to the regular Windows events.

Microsoft's Sysmon, or system monitor, is a Windows system service and device driver that once installed on a system, remains resident across system reboots to monitor and log system activity to the Windows event log. For more information, go to learn.microsoft.com or follow that link.

01:00-02:00
Sysmon adds event IDs of 1-29 and 255. Okay. To install Sysmon on our Windows hosts first of all we download it. From that previous link, there is a download link on that page. We download Sysmon from Microsoft. Open the Sysmon.zip file and copy the Sysmon64 EXE to a download folder or a folder that you at least know where it's being copied to. Then open a command prompt as administrator. Right-click on cmd.exe and run as administrator. Change directories to the download folder or the folder that you chose to put Sysmon64.exe in. Then issue this command. To install Sysmon on your Windows host, the Sysmon64.exe file with the StellarCyber_syswatcher.xml configuration file.

02:00-03:00
We'll use the following command, which is Sysmon64.exe and then accept the end user license agreement. Then the path to the XML file, which is in program data, Stellar Cyber, Windows agents sensor, config, and then the configuration file, the XML file is

stellar_syswatcher.xml. Now of course, for that file to be there, you have to install the Windows Server sensor. If you install Sysmon prior to installing your Windows Server sensor, you won't be able to run this command. You do this afterwards. If you install Sysmon after you've installed your Windows Server sensor software, then we need to restart one of our services. The Windows Agent Sensor Logbeat is the one we need to restart.

03:00-04:00

Open the **Services** app, find our **Windows Agent Sensor Logbeat** service that's running, and right-click and restart or highlight it and then from the left-hand panel, choose **Restart**. Then we want to verify that it's running. Again, in the **Services** app, find Sysmon64 and what we're looking for is that it is running. If you open a command prompt, we can see what the configuration looks like. To verify the Sysmon configuration, open the command prompt as administrator, and execute the command `Sysmon64.exe -c`. That will show you the entire configuration. It will show you the config file that was used. If you scroll down, that screenshot is showing maybe 10 percent of the output from that command.

04:00-05:00

If you scroll down and look, you'll see the entire XML configuration file and the impact on Sysmon. You'll be able to see exactly what's being monitored by Sysmon, which eventually will be put into the Stellar Cyber Data Lake. Let's verify this. Here is a quick way to verify. Navigate to **Investigate | Threat Hunting**. We know by default when we open up **Investigate | Threat Hunting** that the **Alerts** is the index that's chosen, so change the indices or the index from **Alerts** to **Windows Events**. Then have a look in the bottom right-hand side and look for in your Log Class you should now see an entry if you've given it a little bit of time to ingest some of those Sysmon Windows events. You'll now see an entry for Microsoft Windows Sysmon as one of those **Log Source Classes**. Another place to look is again **Investigate | Threat Hunting**. Change the indices from **Alerts**, and select **All**.

05:00-06:10

Now when we select **All**, I want you to go up to the top and type in a search command. The search command we want to look for is **event_ID:**. For searching when we're doing a range, it's an open square bracket, the number 1, capital T-O. It must be a capital. If it's a small t-o, it will not work. Number one, space capital T capital O, space 29. What that's saying is I want to search for event IDs from 1 to 29. Put that in the filter and look for output from that. We can see that there is output. We now have multiple indexes that have been chosen, **Traffic** and the **Windows Events** because some of the Windows events are written to the **Traffic** index. That's another way for us to verify events from Sysmon on the Windows host are being ingested into Stellar Cyber.

The lesson, *Advanced Configuration - Sysmon*, is complete. Thank you.

L08B-DEMO: Advanced Configuration Sysmon (19m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Demonstration of Advanced Configuration - Sysmon. Demonstration of Advanced Configuration - Sysmon.* This lesson includes a demonstration of how to use Microsoft Sysmon system servers to ingest detailed information about process creation network connections and changes to file creation time in addition to the regular Windows events.

In this demonstration, I will show you how to use Microsoft Sysmon system service to ingest detailed information about process creation network connections and changes to file creation time in addition to the regular Windows events. We will install Sysmon with the Stellar Cyber Syswatch configuration file.

01:00-02:00
Following the installation and configuration, we will verify Sysmon events have been ingested into the Data Processor. Let's start with looking at the Stellar Cyber documentation for Sysmon. Let's open up our documentation. There it is down there. Let's browse for it this time. **Installing a Server Sensor** a Windows Server sensor. We're going to scroll down to Installing Sysmon. Nearly there. Okay. Here we are installing Sysmon version 13.31 manually. The first thing we will do is download Sysmon from the Microsoft site using this link.

02:00-03:00
Okay. Here is our Sysmon webpage. The first thing we want to do is download it. We'll click on the Download Sysmon. We can see it's coming down over here. Okay. It opened up a zip. Let's go to the folder. There we have Sysmon. It's a zip file. We have to unzip it. We're going to put Sysmon64 and we're just going to drag that to the downloads folder. Okay. If we go into **Downloads**, there is our Sysmon file that we're looking for.

03:00-04:00
Next, we want to open a command prompt run as administrator. The Windows key cmd, and right-click **Run as administrator**. We want to change to that download folder. There we can see our Sysmon64.exe. That's the file we're after. We want to install that Sysmon64.exe with the Stellar syswatcher.xml configuration file using the following command. Let's go grab that from the documentation. Back to the documentation. Click back there. Here is our command that we're after.

04:00-05:00

Let's copy that and go back to our command prompt and enter. We can see it's installing right now and used that configuration file. The configuration file validated "Sysmon64 installed," the driver installed. It's starting Sysmon. We can see Sysmon has started and then starting Sysmon64 and Sysmon 64 started. That's fantastic! That worked as expected. If we go back to our documentation, we can see down here that restart the winlogbeat service if the Windows Server sensor is already installed, which it is.

05:00-06:00
If we installed Sysmon after the server sensor, which we have, we need to restart the winlogbeat service so that it will work with Sysmon. Let's go ahead and do that. Go to **Server Manager**. **Tools**. **Services**. We know it starts with "w." Select one, press w. There is our winbeat. We're going to right-click, and we're going to restart the service. Alright. Let's now confirm our Sysmon64 installation. First, while we've got the **Services** app open, let's scroll and find Sysmon. Actually I'm just going to type "s" to get to the top of the s's and Sysmon64.

06:00-07:00
There we are. We have it running as a server, so that's a good clarification. Next, let's go back to our command prompt, and we're going to type in the following command: `sysmon64.exe -c`. This is the current configuration of Sysmon. Simply make that a little larger and maybe it will not go on one page. Okay. You can see here, this is the configuration for Sysmon. We can see that it was used with the config file of the program data Stellar Cyber Windows Agent Sensor config, stellarcyber_syswatcher.xml. That's what we're expecting to see.

07:00-08:00
Next, let's open the Windows **Event Viewer**. We'll expand that. From the left-hand panel, we're going to expand **Application and Services Log | Microsoft | Windows**. Let's make this a little larger so we can see. I'm just going to select one and press S to get down to Sysmon. There is our Sysmon. Let's expand that. **Sysmon | Operational**.

08:00-09:00
We can see in the main window we have Event IDs: three, ten. What else have we got? It looks like mainly three and ten. We know that Sysmon has event IDs from one to 29 and 255. We can see that Sysmon is writing new events. We have 495 new events. We can refresh that: 530 events. We can see that the Sysmon source is being added as we speak. Now let's go check on the Data Processor to confirm that these Sysmon events are being ingested into the Stellar Cyber DP. To go and look for those Sysmon events coming into the Data Processor, let's go look in **Investigate | Threat Hunting**.
09:00-10:00

We know it defaults to the **Alerts** filter. Let's change that to **Windows Events**. We could look specifically at our AD3 servers. Let's do that. Let's filter for and we can see over here I can already see down the bottom and the event ID of ten is a Sysmon from one to 29 or all the Sysmon events. Can we see Sysmon over here? There we are Microsoft Windows Sysmon. It's actually the largest class of events coming from AD Server three. Let's filter for. We can see we have in the last 24 hours of all of, which looks like it's come in in the last few moments is 511. Let's shrink that right down to say one hour.

10:00-11:00

Right. There we can see those events coming in. We've got a bunch of Sysmon coming in, confirmation that our configuration on our Windows Server is working with Sysmon. Just out of interest, let's go open up our Interflow dictionary. We're going to search for event_ID, and let's use that to see what events are coming in. We can see we have ten, one, five, four, and sixteen that have come in so far. Our main one being event ID number ten. We could actually go look up what is event ID number ten? If we go back to our documentation and we can scroll down. There is a list of events in here. There is number ten: a ProcessAccess.

11:00-12:00

Let's just do one more test to tie all of this together. We'll start here on our server. We're going to open up a command prompt. We're going to look at Sysmon configuration. That is not how I open it. I want CMD, and we're going to run it with administrator. We're going to do a `sysmon64.exe -c`. This is our configuration. Let's open this up to a larger space. Okay. This is that XML file from Stellar Cyber that shows us our configuration. We can look in here and what we're going to do for this test if we go right to the top here is the beginning of our command. There it is. There's Sysmon64.exe -c shows us our current configuration, which is the configuration file that was used. We're using the stellar_syswatcher.xml.

12:00-13:00

If we look at the rule configuration in here, the one I want to look at is FileCreate. We're going to look at FileCreate. It's going to be a match if it includes but combining a rule using "OR." The "OR" is any of these rules. In the FileCreate we're going to do we're going to put a file into that folder, and we should expect to see a Sysmon event created for that. Let's go ahead and do that. I'll open up **File Explorer** and **This PC | C: drive Windows | System 32**, and **Tasks**. Now **Tasks** isn't there, but we can just go type it in there. It must be a hidden folder.

13:00-14:00

Tasks–there it is. Okay. We're going to create a file in here and we're going to expect to see a Sysmon event generated. Let's do that with Notepad. We're going to create a file with, I don't know, "nasty" in it. We're going to select **File**, **Save as**. We're going to go to this PC

hard drive **Windows | System32 | Tasks**. There it is. Let's call this file, "david.exe" and **Save**. If we go and look at that file, there it is. There's our file that we've just created in there.

14:00-15:00
Let's go see that first of all inside our event viewer. We'll go to our **Server Manager | Tools | Event Viewer**. It's nice and large. We're going to expand our **Applications and Services Logs | Microsoft | Windows**. Instead of "s," I'm going to do "t," and go back up because there is our Sysmon, and we want to find **Operational**. Okay. We have a whole bunch of Syslog events, 1,566 of them in there. Let's see if we can find our david.exe. I can go **Find**, (types) `david`. We have found one.

15:00-16:00
There is our file. The **TargetFilename** is c:\Windows\System32|Tasks\david.exe. That's what's being created in our Sysmon log. If we look at our documentation for Sysmon, Event ID 11 is a FileCreate. We expected to find that there. There it is event ID number 11. Now let's jump across to our DP and see if we can now find this Sysmon event being ingested into Stellar Cyber. Over to our DP, we're going to go do **Investigate | Threat Hunting**. The **Alerts** index is the default. We're going to change that to **Windows Events**.

16:00-17:00
I think we're pretty safe that this happened in the last hour. We know it happened on server AD3, so let's filter for that. We know it's a Sysmon, so we could filter for that. We know it's an Event ID number 11. At the top we could type in `event_ID:11`. There are five of them. Let's add a column down to this table to see the file names that we're looking for. If we scroll down, we can see here is the **Filename**, and we can actually see this is the record. It's david.exe. We're going to add that field. I don't think I clicked on it. Let me do that again. Let me go down here and add the column and there we can see is the file that's been added.

17:00-18:00
Just stretch that out and we can see david.exe. If we look at the details of this record in **Details** we can see it was notepad.exe, which is what we used to create it. That's correct. Clearly it was on AD3. We know that. Our event data target filename, there is our file name. The user that created it, Administrator is what we're currently logged in as and there is our event_ID 11. There's the processor type, our winlogbeat, which is the service that we had to restart after installing Sysmon.

18:00-19:05
We also have our Windows Agent as our message origin source. There it is, full circle for seeing a target file name in our configuration of Sysmon, that anything that goes into that

directory, there'll be a Sysmon event number 11. We've seen that in our **Event Viewer**, and then we saw that ingested into Stellar Cyber.

The lesson, *Demonstration of Advanced Configuration - Sysmon,* is complete. Thank you.

L09 - ADVANCED CONFIGURATION - FILE INTEGRITY MONITORING (FIM)

L09A-PRES: Advanced Configuration - File Integrity Monitoring (FIM) (06m:00s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Advanced Configuration - FIM. Advanced Configuration - FIM.* In this lesson, you will learn how to implement, test, and verify the File Integrity Monitoring feature of the Windows Server sensor.

Another feature of the sensor profile for the Windows Server sensor is that Stellar Cyber includes a File Integrity Monitor feature for Window Server sensors and Linux Server sensors that lets you keep track of changes to specific files and directories on the host server sensor machine. It includes file changes, file creations, and file deletions. You can see in that image that is just the default that comes with Windows. Let's look at how to configure that.

01:00-02:00
First of all, we're going to just set up a test for our File Integrity Monitoring. On the Windows host we're going to go to the c: and the root drive and create a folder called, "FileMonitorTest." That's the folder that we're going to monitor. We'll create it first and then do the configuration inside of the Stellar Cyber UI. Inside the Stellar Cyber UI, we now know that to make a change for a Windows Server sensor, a feature change, we don't do it on the sensor itself. We do this in the sensor profile. For FIM to be configured we go to the S**ensor Profile**. Then it's under the **Windows** tab. First of all, navigate to **System** and **Sensor Profiles** where we can see the profile that we're using: the Training Server Sensor Profile.

02:00-03:00
Click the pencil icon to edit that profile. In the pop-up window we want to select the **Windows** tab because we're doing this for a Windows machine, not a Linux machine. Scroll down. You'll see a slider. It is off by default. You'll see a slider to turn on **FIM Windows**. Select the slider, and you'll see the default pop up of files that will be monitored for change, adds,

and deletes or change, create, and delete. What we want to do is add our FileMonitorTest folder to that. At the top of that table you'll see that there is a **+ Add Row**. Select that. IThat will add a new row to our table. We want to then click in the **Path** cell and add c:\FileMonitorTest. You can see by default the inclusion is a star meaning it includes everything. There are no exclusions.

03:00-04:00

We could create a folder here and exclude certain files from that folder that we would like to monitor, and then we have the depth. Now the depth is how many subfolders we will look down. If the FileMonitorTest folder had subfolder upon subfolder and upon subfolder, we can have a maximum of ten subfolders that we will monitor. After you've done all that, select the **Submit** button to push the change in the Sensor Profile down to our Windows Server sensor host. The next step is for us to test this, to verify that we configured this correctly. Back to our Windows host and in that folder–the c:\FileMonitorTest folder–we're going to create just a test file, any file. You could copy it in there. We just have to create one in this example. It's going to right-click in that folder. Do a new and test file give it a name: testfile is created.

04:00-05:00

After we've got that file created, we should now see, because that folder has been monitored and has been monitored for create, change, and delete, we should see some entries from our Windows Server sensor software sending that off to our Data Processor. Let's go check that in the web UI. Here we are. We're in the user interface. There's a visualization for File Integrity Monitoring. There's a dashboard. If we navigate to **Visualize** and **File Integrity Monitoring**, we're presented with this visualization of pie charts, bar charts, and of course a table. If we look in there, we can see that we now have some add, change, and deletes. Well sorry, not delete, just some create and change for what we've done. The create was actually the very first time the file is created. Then the change is the new file name that's created.

05:00-06:00

We actually see a couple of them in there, but this is where a good place to go find immediately that those File Integrity Monitoring is being sent to our Data Processor. Another place in the user interface is to go to **Investigate | Threat Hunting**, change our index from **Alerts** to **Windows Events**, and start to look for a search of message_class. The message_class is f-i-m. If we look for that, all of those will show up. If you look in the columns in the table, we can see some fields relating to FIM. We can see the file names or the directories that have received records that have triggered in our File Integrity Monitoring feature.

The lesson, *Advanced Configuration - FIM*, is complete. Thank you

L09B-DEMO: Advanced Configuration - File Integrity Monitoring (FIM) (07m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor Installation and Configuration* module is *Demonstration of Advanced Configuration - FIM. Demonstration of Advanced Configuration - FIM*. This lesson includes a demonstration of how to implement, test, and verify Stellar Cyber's File Integrity Monitoring feature.

In this demonstration, I will show you how to implement, test, and verify the FIM, the File Integrity Monitor feature of Stellar Cyber's Windows Server sensor. Let's start with setting up our system with a test folder. We're going to open up **File Explorer**. Let's just go and create a folder in the root of c.

01:00-02:00
We're going to give it a new folder and we're going to call it, "FileMonitor." Okay. We've got c:\FileMonitor. That's what we can see there, right? Okay. Great. Step one. The next step is in the DP UI. Here we need to edit our sensor profile. There are a couple of different ways of getting there. We can either go to our sensor. We know we have a Windows Server sensor on AD3. We can find out that it's that one there. Training Server Sensor Profile. If we click on it there, we edit the profile. That's one way of doing it. The other way to do it is **System | Sensor Profiles**, but I suppose looking at this list is how can I be certain of which profile it is that I'm editing?

02:00-03:00
As we can see, we have a Default standard profile and our Training Server Sensor Profile. But if you had many profiles here and you were unsure of which profile was bound to your Windows Server sensor, the most sure way of doing that is to go to our **Sensor** list, look at our server sensor. That's definitely it. We can see our **SENSOR PROFILE** bound there. A pretty guaranteed way of editing the right sensor profile. Click that, and we are editing the Training Server Sensor Profile. This is a Windows feature, so we choose the **Windows** tab in this screen. If we scroll to the bottom here, we can see **FIM Windows** is not turned on. Nothing to expand there. Let's turn it on, and now we have an expansion that's come out.

03:00-04:00
These are the files that are by default, the files and folders that are being monitored. We can see we've got c:\boot.ini and autoexec.bat and config.sys, and there are two pages. Regedit Explorer, they're the ones being monitored. What we want to do is we're going to add a row. We're going to add in our file monitor, our path, which was c:\FileMonitor, and

Enter. Now we're including everything. We're not excluding anything. A depth of ten is how many subfolders we are going to go down to look for changes, creations, and deletions. That is the configuration piece. Then of course, we've got to go and push that configuration to all of our sensors that are bound to this profile, do that with **Submit**.

04:00-05:00

If we go to our **Sensor** page, we'll probably see, nope, it's happened that quickly. Alright. The configuration's already been pushed to our server. Let's go back to our server and drop in a file in this folder. Let's just go and right-click, select **New**. We'll just put a text document in there. The text document can be latenight.txt. There's our file. We could edit it, put in some test text, save it. We've now made a change to a folder or a directory that we are monitoring on this server. I'll go to **System | Sensors**. We can see our AD3SRV is up and running.

05:00-06:00

Select the refresh button, green light, and this time we can see File Integrity Monitoring has a green light. Let's go ahead and look in **Investigate | Threat Hunting** and look for a message class, not **Alerts**, in **Windows Events**. We're going to look for a message class of FIM. If we scroll down here we can see, we're not seeing it. Remember there's only going to be one. I think we've only edited one, so there will only be one entry. We're looking at 38,000 entries here. Even if we were to say look at the top 25, I'm going to hazard a guess that our one entry doesn't even get a look in in this list of 25. Let's come up here and do a search for MSG_class of f-i-m.

06:00-07:00

There is our one record. Let's go have a look at the details of that. The **Detail** fields of what we're interested in here we can see this was on the computer name AD3SRV. We can see the event owner who was signed in at the time is administrator, that who we are currently signed in as. The module was file integrity and here is our file name. That's the path. The message origin source of our Windows agent. The service type file_integrity, and the source name FIM.

07:00-07:35

We use the message class of FIM for the search. Okay. We can also go and look. There is a visualization. **Visualize** and then **File Integrity Monitoring**. We can see we have one entry here.

The lesson, *Demonstration of Advanced Configuration FIM*, is complete. Thank you.

L10 - ADVANCED CONFIGURATION WINDOWS CHANNEL LOGGING

L10A-PRES: Advanced Configuration - Windows Channel Logging (08m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor* module is *Advanced Configuration - Windows Channel Logging. Advanced Configuration - Windows Channel Logging.* Windows servers can perform a variety of roles. This lesson shows how to enable events relating to those roles and enable the corresponding ingestion through the configuration of channels in the sensor profile.

For Windows installations that include features and roles like IIS Web server or Windows DNS server, you can enable logging for specific channels in Windows and then also in the Sensor Profile. This lesson will go through that process. Looking at IIS on our Windows host, if we open up the **Internet Information Services (IIS) Manager** app, or IIS Manager app.

01:00-02:00
From the **Connections** pane on the left, select the Windows server name, then double-click on the **Logging** icon. From there what we want to change is log events destination to have **Both log file and ETW events** or event tracing for Windows. We want both of those turned on. We're already increasing what IIS will send to our event log in Windows. When you finish doing that, select **Apply**. That first piece of the configuration was IIS itself. Now we're going to go into **Event Viewer** and turn on logging for IIS configuration and IIS logging. To do that, open up the **Event Viewer** app, navigate down to **Applications and Services logs**, and then **Microsoft** and then **Windows**.

02:00-03:00
Expand **IIS-Configuration** and we can see there we have **Administrative** and **Operational**. What we want to do is right-click. We have to do them individually. Right-click on **Administrative** and select **Enable Log**. Next we do the same. Right-click on **Operational** and select **Enable Log**, and then finally expand **IIS-Logging** to reveal the logs folder. There too we will right-click and select **Enable Log**. Now that they're all done, let's go into the **Event Viewer** and actually look for some logs because if we come back to here we can see that the highlighted one: IIS configuration administration, if you look in the right-hand panel at the top it says administration, number of events, zero disabled, because that logging channel has not been turned on yet. That was the reason we needed to come in here and turn all those on.

03:00-04:00

Now that we've turned them all on, if we go back to the **Event Viewer** log, we can see that after a refresh, we now have some logs coming in for that channel. As we say, navigate back to **Applications and Services Logs** | **Microsoft** | **Windows | IIS Configuration** | **Operational**. We can see now inside of that channel we have some logs that have been written to our **Event Viewer**. Note that there are two events in there. There is an event for 50 and event ID number 29. Just keep those two numbers in mind: 50 and 29. We're going to look for those inside the UI of the Stellar Cyber Data Platform. Here we are inside our DP. Now turning all that on for IIS for logging and then going to the **Event Viewer** and turning on logging for those channels, is just the first step.

04:00-05:00

The second piece is to tell our Windows Server sensor to forward those logs to go inquire for those logs, pull them from **Event Viewer**, normalize them, and then send them to our DP. To do that, of course, we're going to go into, it's not our server sensor that we're going to do it, we're going to go make a configuration change to our sensor profile. Navigate to **System | Sensor Profiles**, click the pencil to edit the profile. The **Windows** tab is where we will do this. You can see by default, there are a number of tabs and features and channels that have already been turned on. But the ones we're looking for are quite specific. You'll see there's actually quite a long list of channels that are available to turn on. If you go down and into other channels and open that, there is now a box that you can click in. When you click in that box, you'll see that there is a very long list of Windows channels that we can open.

05:00-06:00

Rather than scrolling through them all, we can type in `IIS` and we can see the three IIS channels appear. They're the three channels that we enabled logging for in our **Event Viewer**. One by one, check the box to enable it. We can't check them all at one time. We select them one at a time, and then go back in, and type `IIS`. The two remaining will come up. Check one, and then type in `IIS` again. The remaining one will appear. Check that. Then all three have been checked. Once that's been done, we select **Submit**, which will change the configuration of the server sensor profile and will be pushed down to our Windows host to make that change. To validate that configuration, let's go in to **Investigate | Threat Hunting**. You can see we're going to change the indices from **Alerts** to **Windows Events**.

06:00-07:00

What I want you to pay attention to is on the right-hand side down in our Log Classes that we now have additional classes in there. We have Microsoft Windows IIS Configuration, Microsoft Windows IIS Logging. You can see that there are two new channels in there for which we are receiving Windows events for. Staying with **Investigate | Threat Hunting**

another way we could go about seeing this is to put in a search term in the search bar: event_ID: 29 OR event_ID: 50. What that does, opposed to that previous search we did where we were looking for a range, this is saying I specifically want to see event IDs 29 and 50. Remember, if we mapped back to what we saw in the event viewer that that channel was open showing us those two events. This is another way we could do that search.

07:00-08:05
If we look in the table below, we have an event ID of both 29 and 50 and then we're seeing another column for message class showing us Microsoft-Windows-IIS-Configuration. That's the channel that we saw inside of our **Event Viewer**. There is full circle, enabling more detailed logging in the IIS role for that server, then enabling logging on those channels, then again enabling those roles on those channels inside of **Event Viewer**. On the Stellar Cyber side going into the sensor profile and enabling those IIS channels. Remember, we enabled those three channels? This is the result of what we're seeing. Now we're ingesting IIS traffic from our Windows host.

The lesson, *Advanced Configuration - Windows Channel Logging*, is complete. Thank you.

L10B-DEMO: Advanced Configuration Windows Channel Logging (16m:00s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor* module is *Demonstration of Advanced Configuration - Windows Channel Logging. Demonstration of Advanced Configuration - Windows Channel Logging*. Windows servers can perform a variety of roles. This lesson includes a demonstration of how to enable events relating to those roles and enable the corresponding ingestion through the configuration of channels in the sensor profile.

In this demonstration, I will show you how to add additional Windows Events based on event logging from additional event channels. I will step through the configuration required of the Windows Server and the corresponding steps for the sensor profile in the Stellar Cyber Data Processor. Finally, we will validate the ingestion of these new events. Let's start with looking at the configuration of IIS on the Windows server.

01:00-02:00
Alright. Here we are in our 2019 Windows server. It's a fresh install, so it does not have IIS running. I will just walk through the procedure quite quickly. We'll speed up the video just so you can see that we're going to install that piece from scratch. Standby while we do that.

02:00-03:00
Back in our DP, let's look for those IIS Windows events coming in. **Investigate | Threat Hunting**. We're looking at the last 24 hours. Let's unselect **Alerts**, and then select **Windows Events**. We're going to filter for our server. Let's just have a look. There are lots of them. Remember we had this issue with File Integrity Monitoring that maybe they are coming in, but they're so small that also few of those that they don't make the top 25.

03:00-04:00
We can go up here and let's do a search for msg_class: iis. We'll see we don't have any of those in there. Just to confirm that that is the correct search, we'll clear our search. Let's do a filter for. Let's do it for Sysmon and we can go into our **Filters** section. We can come down. We can see we are filtering for Sysmon, copy as query. We can paste that. Let's close that now. Come up to the top and paste that query here. We can see we are looking for msg_class: We could get rid of really all of this and just put in Sysmon. We can see there that that will show up. If we change from Sysmon to IIS, we have zero.

04:00-05:00
Back we go to the Windows Server to turn on logging for our Sysmon channels, then we'll come back to our Data Processor and make those changes to the Sensor Profile. Here we are. Let's spin up IIS. **Tools | Internet Information Services (IIS) Manager**. We'll expand that. From the left-hand pane we're going to choose our server that's running IIS. Open up **Logging**. We're just going to add **Both log file and ETW event**. ETW is event tracing for Windows, so a little extra information we're expecting to come through. Select **Apply**. The changes have been successfully saved. The next thing we need to check is that logging is enabled for IIS-Configuration and IIS-Logging.

05:00-06:00
We do that in the **Event Viewer**. We can minimize that. **Tools | Event Viewer**. We navigate to **Applications and Service Logs | Microsoft | Windows** and we're going to scroll down to find **IIS**. Why don't we choose one and press "i"? There we go. There are the two we're looking for: **IIS-Configuration** and **IIS-Logging**. If we expand those out, well, actually let's just highlight Configuration. You can see there zero events. Administrative is disabled. We right-click, and select **Enable Log**.

06:00-07:00
It's the same for Operational. It's disabled. Right-click, select **Enable Log**. Let's do the same for **IIS-Logging**. Zero events, **Logs**, disabled, right-click, **Enable Log**. Let's make some changes to IIS two for some logging. Why don't we do something with authentication? Authentication for **anonymous** is on. Let's disable that. Well, maybe we decide we want to

enable again. We'll enable again and check. Let's go back and check our logs to see if we have any additional logs. That's in **Operational**. There we go. We can see we have some changes that have been committed there: 50 and 29.

07:00-08:00

We can see there: authentication anonymous successfully being committed. A couple of changes now. We're seeing some logs in **Administrative**. We have some warning logs in there too, so IIS is throwing out some events. Let's go into our Data Processor and see those events in there. Back in our Data Processor, **Investigate | Threat Hunting**. Change **Alerts** to **Windows Events**. Sorry. Let's uncheck **Alerts** so we've just got **Windows Events**, and let's go back to just looking at our server. Filter for, we can see in there where for our sensor we're only looking at our Windows Server sensor. Do we have? We don't know. It might be too hard, right?

08:00-09:00

Too small being that in the last 24 hours there are 50,000 events coming from that server now that we turned it on to the highest volume of logging in our sensor profile. Let's just check for `msg_class: iis`. Zero. Why is that? Why have we got zero? Well, there's still one more configuration item left to do and that is change our profile. Remember, we can go into our profile. We know what it's called. It is this one but if we were unsure and we were expecting from **Investigate | Threat Hunting**, **Investigate | Threat Hunting**. If we knew we were expecting to see IIS class of logs coming in from our server, zero, and we know that we're looking at our server. Our server is AD3SRV. We could go to **System | Sensors**.

09:00-10:00

It's a Windows Server sensor. That's what we're looking for and we can see there it's running Windows and AD3SRV is using the training server. I can't quite see it all: Training Server Sensor Profile. If we click on that button, now we're editing that profile. The one last thing to do is–this is a Windows function. Select the **Windows** tab, scroll down. This is not a template issue. This is a channel issue. Other channels, and we can type in some other channels. When I click into that field, all of these come up. I'm just going to search for IIS. There are the three that we're looking at: Configuration Administrative, Configuration Operational, and Logging Logs. If we go back to our server, we can see that those were the ones we're looking for: IIS Configuration for Administrative and Operational, and IIS Logging for logs.

10:00-11:00

Back to our DP and there are the three. Now I can't select them all in one go. I chose one. I type in `IIS` again, I select the next one `IIS`, and I choose the last one. Effectively now we're adding those channels to our server sensor software to bring in events that match those channels. The last thing to do is select **Submit** to push that new sensor profile. We can see

immediately our sensor has gone red. It is loading the new configuration of our sensor profile in that server sensor software, not rebooting the host at all, just restarting our server sensor software on that Windows host. Let's do a refresh. Green up and running again. My expectation now is to see some events coming in with a message class containing IIS. Let's go have a look at that.

11:00-12:00

Investigate | Threat Hunting | Alerts. Here's one thing that's remained the same. Even though we went into **System** and **Sensors** and **System** and **Sensor Profiles**, our filters remained persistent. We're still filtering for AD3SRV and we still have our search up here msg_class: IIS. But the default whenever we're going to **Investigate | Threat Hunting** is **Alerts**. Let's select change that back and **Windows Events**, and look at that. We have Microsoft Windows IIS configuration. They're the first of our events coming through under the banner of IIS. Going back to our server, let's just see what they are. We can see in here these Operational ones are 29 and 50. Back to our DP: 29 and 50.

12:00-13:00

We're also seeing a 12 in there. There's probably also a 12 listed in this list somewhere. I could filter the current log and filter just for 12. Let's see if there's some in there. My mistake. The 12 are under **Administrative**. The channel that we're looking at here is Microsoft Windows IIS Configuration, which is this entire. It's the bundle of those two we're seeing that's coming in. Let's look at the **Operational** one. We'll just clear our filter over here. These were our changes that we made about our anonymous authentication. This was event ID number 29. Let's go filter for those event ID 29 and see if we can see this log in our DP that's been normalized, enriched, and ingested.

13:00-14:00

Number 29: If we do a right-click there and select **Add as Including Filter**, we'll only see 29. Now we have six of those in the last 24 hours. What are we on? In the last 24 hours. Hit our **More Info** button, and let's look at some of these details. Computer_name is the name of our host that we're working with. The engine_ID name, that is the server sensor software. Under event data.configuration we can see that change that we have enabled anonymous authentication. There's our event ID, number 29. Our two pieces of information here–our log name is Microsoft IIS Configuration Operational.

14:00-15:00

That's a little sub of Microsoft IIS Configuration. Remember under Operation, sorry under Configuration we had Operational and Administrative. That's where we'll see both of those show up. That's how we add additional Windows Event channels to our sensor profile. Not only do we need to turn on that logging information on the Windows server so that the

events produced, we also need to tell our Windows Server sensor software to pull in those logs, normalize, enrich, and send them to our Data Processor. We do that via the Sensor Profile as we saw. We know under Microsoft Windows IIS Configuration we have Administrative and Operational. Let's look at those. We can see that they're in here–the 59 sorry the 29 the 50. Let's go and add those in there to see where they are.

15:00-16:00
We need to find Operational. There it is. It's the **Log Name**. We show column. Now I can see that column there. Let's collapse this record. Give it a bit of space so we can see the end of it and then we can see it. 29 is Operational. 12 is Administrative. We know that 29 and 50 were under Operational and 12 is under Administrative. That's where we can see that connection of one: turning on IIS, two: enabling logging, three: enabling the event viewer to enable those channels of logs, and then finally to change the Sensor Profile to collect those logs from the server and bring them into our Data Processor.

The lesson, *Demonstration of Advanced Configuration - Windows Channel Logging*, is complete. Thank you.

L11 - TROUBLESHOOTING

L11A-PRES: Troubleshooting (08m:00s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor* module is *Troubleshooting. Troubleshooting*. We sometimes make errors during the installation and configuration of the Windows Server sensor software. This lesson looks at how to identify those errors and rectify some of them.

To finish off this final piece of our Stellar Cyber Windows Server sensor, let's look at a couple of troubleshooting and changes we might make in relation to IP addressing of our Windows Server sensor. During the installation of that software we're asked to put in an IP address, as we can see here. The example for this one is that network one or that first network configuration where we have a Windows Server sensor communicating through a firewall directly to our Data Processor. We've been using 192.168.1.175.

01:00-02:00
What would happen if you actually made a configuration error or a typo, you "fat fingered" something when putting that IP address in and you finish your installation, go back to our web UI, look in the interface **System | Sensors** expecting to see our newly installed Windows

Server sensor, and all we see is a blank page. There are no sensors here. We're expecting to see it with a green dot that the communication is running. Fair enough, that we need to authorize it. There'll be that red shield, but instead we don't see anything here at all. What can we do? One of the things we do and let's do this before we actually get to this configuration, let's say we open up our Windows CLI app in Windows, jump in and do a `show cm`.

02:00-03:00

That would show us that we've what our configuration file is for our Windows Server sensor. If that IP address is wrong, this is how we go about changing it. We navigate to c:\Program Files\Aella\, and we'll see in there that there is a file, a shortcut file called **config**. If you double-click that **config** file, you'll be brought to c:\ProgramData\StellarCyber\WindowsAgentSensor\config. That's the folder we'll end up in, or the directory that we'll end up in. What we need to do is look for a file called **aos.yaml**. We need to find that and edit that with a text editor. We see what we've got in the screenshot on the right-hand side is the configuration of our Windows Server sensor. We're going to search for our IP address and correct it. Maybe we had 192.169.1.175.

03:00-04:00

We saw the error. This is where we go and reconfigure it. We don't want to have to uninstall the software and reinstall it. We just need to make that configuration change. Make the configuration change with your text editor, and then save the file. After that's been done, we need to tell the server sensor software that we've made a change to the configuration file and we need to reload it. The way we do that is to open up the **Services** app on the Windows host and restart Windows Agent Sensor Conf. That's the service that needs to be restarted. You can either highlight the service and from the left-hand panel click **Restart** or you can right-click on the service and restart from there. After that's done, it will reload our aos.yaml file with the correct IP addressing and we'll be good to go.

04:00-05:00

The other thing we could do, so we know we've been working with this configuration of a Windows Server sensor on the left-hand side. On the right-hand side, we've had another network. 10.20.1 network on the left-hand side of this network. On the right-hand side, 192.168.1 is network on the right-hand side .175 for the Data Processor. If we were to change that architecture and say, “I've got too many server sensors going through that firewall, I want to change this to insert an Aggregator.” We could go with what we saw at the very beginning a different network architecture, where we have a Modular Sensor between our Windows Server sensor or a Linux Server sensor or other Modular Sensors to communicate to a Modular Sensor with the Aggregator feature turned on. It's that Modular Sensor with the Aggregator feature turned on that communicates directly with the Data Processor.

05:00-06:00
Our environment, we have a Modular Sensor on the left-hand side. There's actually no left-hand firewall in the configuration that we've been using. We've got a Windows Server sensor on the 10.20.1 network and we have a Modular Sensor on the 10.20.1 network. What we're going to do is look at how we would change that configuration of our Windows Server sensor communicating to 192.168.1.175 directly and instead have it go via our Modular Sensor Aggregator. To do that, I know we've been changing everything in the sensor profile and then pushing it out to any sensors that are bound to that profile, but in this instance we actually edit the sensor itself. We will go to **System** and then **Sensors**, we will find our sensor in that table and click on the pencil to make that edit.

06:00-07:00
After we get the pop-up window the edit sensor parameters pop up go down and expand on the primary Aggregator expansion and we can see there we have either none or we can choose. Now the available sensors will be ones that have the Aggregator feature turned on. In this example, sc-modular-sensor has that Aggregator feature turned on. Select that. When you're finished, click **Submit**. That will push the new configuration down to our Windows Server sensor, and now it will communicate via a Modular Sensor. How do we check that? Jump back onto our Windows host, open up the Windows Agent sensor CLI app, and then issue the `show cm` command. We can see if we do that. I want you to notice that the CM controller IP address has not changed. This server sensor is still communicating to that Data Processor: the 192.168.1.175.

07:00-08:00
What has changed is there's now a next hop address in there of 10.20.1.15. That is our Modular Sensor. The other command that you can run is the `show aggregator` command, and what I want you to verify in the output of that command is we can now see that there is a primary Aggregator of 10.20.1.15, that Modular Sensor sc-modular-sensor. We can see the status is up. That's how we would change from a network architecture where we have our server sensors communicating directly with our Data Processor to a network architecture where we have our sensor communicate via a Modular Sensor with the Aggregator feature turned on.

The lesson, *Troubleshooting*, is complete. Thank you.

L11B-DEMO: Troubleshooting (11m:15s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Windows Server Sensor* module is *Demonstration of Troubleshooting. Demonstration of Troubleshooting*. This lesson includes a demonstration of how to identify and rectify errors made during the installation and configuration of the Windows Server sensor software.

In this demonstration I will show you how to make changes to the Windows Server sensor configuration in case you make a mistake or need to make changes after the installation. Let's start with looking at the installation configuration of the Windows Server sensor. When we configured it, when we installed the Windows Server sensor software, we added our CM address and we added our address of 192.168.1.175.

01:00-02:00
But when I go to my UI and I go to **System | Sensors** it's not showing up. I've waited and waited and I've refreshed the screen. I've refreshed this table and my Windows Server sensor is not appearing for me to authorize and continue the configuration. If I go to my Windows host let's look at the services running. Navigate to **Tools**, and then **Services**. I know it starts with "w." Here we are. Running, running everything looks fine, all running and happy. Why is it that I cannot see my Windows Server sensor in the sensor table in the UI? Let's run the CLI command. There are a couple of ways you can get here. This is just Windows. I can press the Windows button and search for it, Windows Agent Sensor CLI,

02:00-03:00
or you just do Windows, type `cli`, and it will show up. Here we are in the CLI of our Windows Server sensor software. If we do a `show cm`, I have 192.168.1.176. The address I'm looking for, if I go into my UI, is **System | Data Lake**, and I am looking for this address: 192.168.1.175. I have the incorrect address. During the installation, I typed the wrong address here. How do I go about changing that? Back to the Windows host. There's a configuration file that I need to edit. If I open File Explorer,

03:00-04:00
and select **This PC**, C drive program files, Aella, and this shortcut config takes us to a new folder. It's **aos.yaml** that I need to edit. Right-click. Select **Edit**. Just the text editor is fine. Let's expand that out. Right. There's my error. It needed to be 175. I make that change, close the file, and save it. But I've got to go and restart the server. Let's go to our **Services** and I restart the configuration service. Select **Restart the service**. Let's wait for that to restart there. I can go back to my CLI. It'll be a period of time before that `show cm` works, but let's just have a look. There we go! Straight away, `show cm` 192.168.1.175 and the **Status** is UP.
04:00-05:00
That's great news. Let's go over to our DP and see if we can see it in there now. Here we are in our DP, **System | Sensors** and I have my AD3 server. All I need to do is authorize that

server now. Select **Manage | Authorize**, add it to the list. Select **Add**, and then select **Submit**. Let's see if that's happily up and running now. Refresh, not yet. We can go to our server check with the CLI. Nope not running yet. We're still pushing that config down to this server sensor. It's starting to come up there. We can see 192.168.1.175 but **Status** is still DOWN. Let's just check again.

05:00-06:00

There we go, up and running back to our sensor list, refresh. It won't be long now. It's green, all happy. The one change we do want to make though is because we've just added it, it's back to our default, and we don't want to use that. We created one earlier. Let's change that and **Submit**. Again, we have to wait for the server sensor profile that we've just added called Training Server Sensor Profile is being pushed down to our Windows Server sensor. Up and running, go check it out on the sensor. All happy there. Now looking at our two different network architectures, the current one we have is we have our Windows Server sensor there on the left.

06:00-07:00

At the moment we know that's at 10.20.1.13. I don't know that I've ever shown you that. Let's just go have a quick look there. CMD and we want to do an `ipconfig` 10.20.1.13. We can also see that in our DP list, sorry our DP list on our Data Processor, we can see the server here is 10.20.1.13. Currently communicating directly with our Data Processor through a firewall and of course we have to have all of these firewall ports open for that communication to happen, but if I had a different architecture, if I had an architecture that looked more like this, where I have a Windows Server sensor behind the firewall, talking to a Modular Sensor with the Aggregator feature turned on, that is also going through a firewall to the Data Processor. If that was our network architecture there's no reason why we still couldn't have our Windows Server there on the left of 10.20.1.13

07:00-08:00

communicating directly to our Data Processor, that 192.168.1.175, but that would require me to have firewall ports open for all my different Windows Server sensors, Modular Sensors, Linux Server sensors, in that right-hand firewall open. We don't necessarily have to have that left-hand firewall in this picture for the scenario we're going to do, but instead of the Windows Server sensor communicating directly to the Data Processor, we're going to enable aggregation so that the Windows Server sensor will communicate to the Modular Sensor with an Aggregator feature turned on. That will directly communicate with the Data Processor. Let's see how that would look for us to configure that. Okay. First, we'll look at our architecture here.

08:00-09:00

We've got our server on 10.20.1.13 communicating directly to that 192.168.1.175. We have a Modular Sensor here, which is 10.20.1.15. These guys are on the same network–that Modular Sensor and that Windows host. Let's have the Windows Server sensor communicate via the Aggregator, the Modular Sensor to the Data Processor. Let's look at how we would configure that. The first thing to do is edit the server sensor. We can see here we have a Primary Aggregator. Let's change that. There's nothing to change to. There's no options available, so that's not going to work for us. I wonder why that is? Let's check out the settings for our Modular Sensor. Let's look at the Modular Sensor.

09:00-10:00

We can see here that it's bound to the profile SC Modular Sensor New. If I click on that button we open up editing of that sensor profile. We can see that the Aggregator feature is not turned on. Let's turn that on, and then select **Submit**. While we're doing that, that configuration of this that we just changed in the profile is being pushed down to the sensor. If we go have a look at the sensor table, we will probably see a big red mark, which we do, for the status of the Modular Sensor. Let's just wait for that configuration to be in sync, not out of sync. There we go. We are ready to go. If I go back to our Windows Server sensor, I edit Primary Aggregator. There it is available. I select that, and then select **Submit**. What's happening now is this device is having its configuration updated and being pushed down to that server.

10:00-11:15

We'll just wait for that to occur, and then we'll go confirm the configuration has been set. Alright. We've got a green light there. Let's go to our server host to have a look. Alright. Here we are. Enter `show cm`. We can see now we have a CM controller still communicating to the same Data Processor, 192.168.1.175, but we now have a next hop of 10.20.1.15, or our Modular Sensor that has that Aggregator feature turned on. We can do a `show aggregator` as well, are just a different command showing us additional information. We can see there our Primary Aggregator 10.20.1.15 is UP and our CM controller 192.168.1.175 is UP. Everything is working as expected.

The lesson, *Demonstration of Troubleshooting*, is complete. Thank you.